C Programming
Pattern Printing Practice

Placement Coding Practice - Programs, Output and Loop Logic

This booklet is intended for classroom and placement practice. Each program accepts the size from the user so that students can test the logic with different inputs.

How to practise

- First draw the expected output for n = 4 or n = 5.
- Identify what changes row by row: stars, numbers, letters or spaces.
- Write the outer loop for rows and then decide the inner loop conditions.
- After coding, test with a small value before trying a larger input.

Questions 1-12 correspond to the supplied practice set. Questions 13-17 are additional exercises for placement preparation.

1. Right Triangle - Star Pattern

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
*
* *
* * *
* * * *
* * * * *
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }

    return 0;
}
```

Logic: Row number decides how many stars are printed.

2. Inverted Right Triangle

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
* * * * *
* * * *
* * *
* *
*
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }

    return 0;
}
```

Logic: Start with n stars and reduce the count by one in every row.

3. Pyramid Pattern

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
    *
   ***
  *****
 *******
*********
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space;
```

```
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");

        for (j = 1; j <= 2 * i - 1; j++)
            printf("*");

        printf("\n");
    }

    return 0;
}
```

Logic: Each row contains n-i leading spaces and 2*i-1 stars.

4. Inverted Pyramid

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
*********
 *******
  *****
   ***
    *
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");

        for (j = 1; j <= 2 * i - 1; j++)
            printf("*");

        printf("\n");
    }

    return 0;
}
```

Logic: The number of stars decreases by two while leading spaces increase.

5. Diamond Pattern

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space;

    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");
        for (j = 1; j <= 2 * i - 1; j++)
            printf("*");
        printf("\n");
    }

    for (i = n - 1; i >= 1; i--)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");
        for (j = 1; j <= 2 * i - 1; j++)
            printf("*");
        printf("\n");
    }

    return 0;
}
```

Logic: Print a pyramid first, followed by an inverted pyramid without repeating the middle row.

6. Hollow Square

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
* * * * *
*       *
```

```
*       *
*       *
* * * * *
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter size: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n; j++)
        {
            if (i == 1 || i == n || j == 1 || j == n)
                printf("* ");
            else
                printf("  ");
        }
        printf("\n");
    }

    return 0;
}
```

Logic: Print a star only when the current position belongs to the border.

7. Floyd's Triangle

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, count = 1;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
        {
```

```
            printf("%d ", count);
            count++;
        }
        printf("\n");
    }

    return 0;
}
```

Logic: Maintain one counter and continue it across rows.

8. Number Triangle - Row Number Repeated

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", i);

        printf("\n");
    }

    return 0;
}
```

Logic: Print the current row number i exactly i times.

9. Pascal's Triangle

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space;
    long value;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        for (space = 0; space < n - i - 1; space++)
            printf(" ");

        value = 1;

        for (j = 0; j <= i; j++)
        {
            printf("%ld ", value);
            value = value * (i - j) / (j + 1);
        }

        printf("\n");
    }

    return 0;
}
```

Logic: Generate each binomial coefficient from the previous value; no factorial function is required.

10. Butterfly Pattern

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
*          *
**        **
***      ***
****    ****
**********
**********
****    ****
***      ***
**        **
*          *
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space;

    printf("Enter number of rows for half: ");
```

```
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
            printf("*");

        for (space = 1; space <= 2 * (n - i); space++)
            printf(" ");

        for (j = 1; j <= i; j++)
            printf("*");

        printf("\n");
    }

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("*");

        for (space = 1; space <= 2 * (n - i); space++)
            printf(" ");

        for (j = 1; j <= i; j++)
            printf("*");

        printf("\n");
    }

    return 0;
}
```

Logic: The two star blocks grow toward the centre and then shrink.

11. Alphabet Triangle

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
A
A B
A B C
A B C D
A B C D E
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;
    char ch;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
```

```
    {
        ch = 'A';

        for (j = 1; j <= i; j++)
        {
            printf("%c ", ch);
            ch++;
        }

        printf("\n");
    }

    return 0;
}
```

Logic: Reset ch to 'A' for every row and increment it inside the row.

12. Hollow Diamond

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space, width;

    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");

        width = 2 * i - 1;

        for (j = 1; j <= width; j++)
        {
            if (j == 1 || j == width)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
```

```
    for (i = n - 1; i >= 1; i--)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");

        width = 2 * i - 1;

        for (j = 1; j <= width; j++)
        {
            if (j == 1 || j == width)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }

    return 0;
}
```

Logic: Only the first and last positions of each row width are printed as stars.

13. Half Number Pyramid - Increasing Numbers

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", j);

        printf("\n");
    }

    return 0;
}
```

Logic: Unlike the repeated-row pattern, this pattern prints the column number j.

14. Reverse Number Triangle

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", j);

        printf("\n");
    }

    return 0;
}
```

Logic: The outer loop controls the decreasing row length.

15. 0-1 Triangle

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
        {
            if ((i + j) % 2 == 0)
                printf("1 ");
            else
                printf("0 ");
        }

        printf("\n");
    }

    return 0;
}
```

Logic: Parity of i+j decides whether 1 or 0 is printed.

16. Palindromic Number Pyramid

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
    1
   212
  32123
 4321234
543212345
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");

        for (j = i; j >= 1; j--)
            printf("%d", j);

        for (j = 2; j <= i; j++)
            printf("%d", j);

        printf("\n");
    }

    return 0;
}
```

Logic: Split each row into a descending part and an ascending part.

17. Hollow Pyramid

Task: Write a C program to print the following pattern for a suitable value of n.

Expected pattern

```
    *
   * *
  *   *
 *     *
*********
```

C program

```
#include <stdio.h>

int main()
{
    int n, i, j, space, width;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (space = 1; space <= n - i; space++)
            printf(" ");

        width = 2 * i - 1;

        for (j = 1; j <= width; j++)
        {
            if (i == n || j == 1 || j == width)
                printf("*");
            else
                printf(" ");
        }

        printf("\n");
    }

    return 0;
}
```

Logic: The last row is completely filled; other rows contain only two boundary stars.

Practice Variations

1. Modify any star pattern so that it prints numbers instead of stars.
2. Modify the alphabet triangle so that every row starts with the row's corresponding letter.
3. Print the diamond using numbers and then using alphabets.
4. Write the hollow square program for separate length and breadth values to form a rectangle.
5. For any two patterns, write functions and call them from main() instead of writing all logic directly in main().

Placement Round Reminder

For pattern questions, concentrate on loop bounds. The outer loop normally controls rows. Inner loops control leading spaces and the symbols printed in each row. For symmetric patterns, solve the upper and lower portions separately. Dry-run the program for n = 3 or n = 4 before finalising the code.